12.2 Independent and Dependent Events

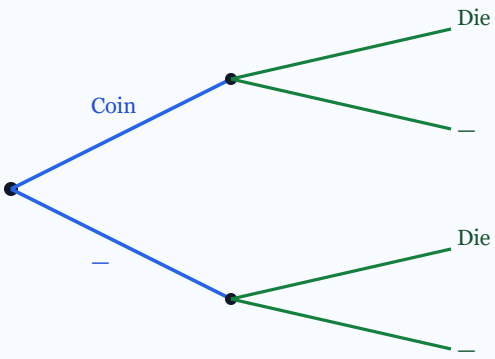
Two events are **independent** if the outcome of one has no effect on the probability of the other (a coin flip doesn't change a die roll) — then $P(A \text{ and } B) = P(A) \cdot P(B)$, using the *same* probability for B regardless of A. Two events are **dependent** if one outcome changes the probability of the next — the classic signal is drawing **without replacement**, where removing an item changes the totals for the next draw. This worksheet practises multiplying along both types of chains, recognizing which type a situation is, and extending to three or more sequential events.

Examples

Example 1: Independent events — coin and die

A coin is flipped and a die is rolled. Find $P(\text{heads and a } 6)$.

Solution



Independent events: the coin's outcome doesn't change the die's probabilities on either branch.

The coin and die don't affect each other, so multiply the individual probabilities directly:

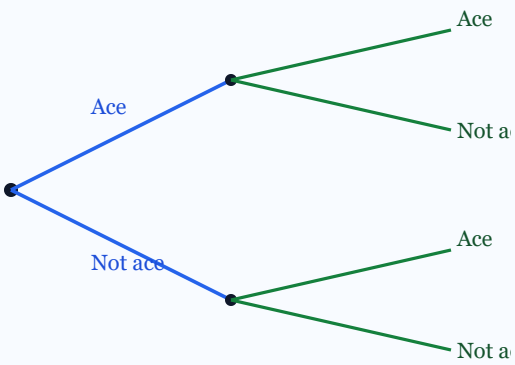
$$P(\text{heads}) \times P(6) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}.$$

Answer: $\frac{1}{12}$.

Example 2: Dependent — cards without replacement

Two cards are drawn from a 52-card deck without replacement. Find $P(\text{both are aces})$.

Solution



Second-draw probabilities depend on whether an ace was already removed — a dependent chain.

First draw: 4 aces out of 52 cards: $P(\text{ace}_1) = \frac{4}{52}$.

Second draw (one ace already removed, one fewer card total): $P(\text{ace}_2 \mid \text{ace}_1) = \frac{3}{51}$.

Multiply along the chain: $P(\text{both aces}) = \frac{4}{52} \times \frac{3}{51} = \frac{12}{2652} = \frac{1}{221}$.

Answer: $\frac{1}{221}$.

Example 3: Marbles — dependent (without replacement)

A bag has 4 red and 6 blue marbles. Two are drawn without replacement. Find $P(\text{both blue})$.

Solution

First draw: $\frac{6}{10}$.

Second draw (one blue removed — now 5 blue out of 9 total): $\frac{5}{9}$.

$P(\text{both blue}) = \frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \frac{1}{3}$.

Answer: $\frac{1}{3}$.

Example 4: Same bag — with replacement (independent)

Same bag (4 red, 6 blue), but each marble is put back before the next draw. Find $P(\text{both blue})$.

Solution

With replacement, the bag resets, so both draws use the same probability — this makes the events **independent**:

$$P(\text{both blue}) = \frac{6}{10} \times \frac{6}{10} = \frac{36}{100} = \frac{9}{25}.$$

Answer: $\frac{9}{25}$. *Compare to Example 3 ($\frac{1}{3} \approx 0.333$) vs. here ($\frac{9}{25} = 0.36$) — replacing slightly raises the probability, since the pool never shrinks.*

Example 5: Identify independent vs. dependent

State whether each is independent or dependent, with a one-sentence reason: (a) rolling a die twice; (b) drawing two cards without replacement; (c) drawing two cards with replacement.

Solution

(a) Independent — the first roll doesn't change the die or its probabilities for the second roll.

(b) Dependent — removing the first card changes both the count of remaining cards and possibly the count of the target card.

(c) Independent — replacing the card resets the deck to its original state before the second draw.

Answer: (a) independent, (b) dependent, (c) independent.

Example 6: Two dice — independent

Two fair dice are rolled. Find $P(\text{both show } 1)$.

Solution



Each die independently favors outcome 1; multiply the two separate probabilities.

Each die is unaffected by the other:

$$P(1 \text{ and } 1) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}.$$

Answer: $\frac{1}{36}$.

Example 7: Cards — different ranks, without replacement

Two cards are drawn without replacement. Find $P(\text{king, then queen})$.

Solution

First draw (king): $\frac{4}{52}$.

Second draw (queen — total cards drop to 51, but queens are untouched since a king was removed, not a queen): $\frac{4}{51}$.

$$P(\text{king then queen}) = \frac{4}{52} \times \frac{4}{51} = \frac{16}{2652} = \frac{4}{663}.$$

Answer: $\frac{4}{663}$.

Example 8: Coins — heads then tails

Two fair coins are flipped in order. Find P(heads then tails).

Solution

Independent flips: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

Answer: $\frac{1}{4}$. *Note this is different from "exactly one head" (Lesson 12.1), which also counts tails-then-heads.*

Example 9: Marbles — three colors, without replacement

A bag has 3 green and 5 yellow marbles. Two are drawn without replacement. Find P(both green).

Solution

First draw: 3 green out of 8 total: $\frac{3}{8}$.

Second draw (one green removed — now 2 green out of 7 total): $\frac{2}{7}$.

Multiply along the chain: $\frac{3}{8} \times \frac{2}{7} = \frac{6}{56} = \frac{3}{28}$.

Answer: $\frac{3}{28}$.

Example 10: Three-stage dependent chain

Three cards are drawn from a 52-card deck without replacement. Find P(all three are aces).

Solution

Chain three dependent draws, each removing one ace and one card from the totals:

$P = \frac{4}{52} \times \frac{3}{51} \times \frac{2}{50}$.

Numerator: $4 \times 3 \times 2 = 24$. Denominator: $52 \times 51 \times 50 = 132{,}600$.

$P = \frac{24}{132{,}600}$. Divide both by 24: $\frac{1}{5525}$.

Answer: $\frac{1}{5525}$.

Example 11: Word problem — quality control (independence assumption)

A factory's defect rate is 2% ($P(\text{defective}) = 0.02$). Two items are chosen at random from a very large batch. Find $P(\text{both are defective})$.

Solution

Because the batch is very large, removing one item barely changes the defect rate for the next — the events are treated as (approximately) independent, so $P(\text{defective}) = 0.02 = \frac{1}{50}$ for each pick.

$$P(\text{both defective}) = \frac{1}{50} \times \frac{1}{50} = \frac{1}{2500}.$$

Answer: $\frac{1}{2500}$ (or 0.0004).

NAME: _____ Geometry (FL B.E.S.T.)

Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Independent

A coin is flipped and a die is rolled. Find P(tails and even).

Answer: **Verification:**

Problem 2: With replacement

Bag: 5 red, 5 blue. With replacement, find P(both red).

Answer: **Verification:**

Problem 3: Without replacement

Same bag. Without replacement, find P(both red).

Answer: **Verification:**

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Problem 4: Identify

Two rolls of a die: independent or dependent?

Answer: **Verification:**

Problem 5: Cards

Without replacement, find P(two hearts).

Answer: **Verification:**

Problem 6: Dice

Two dice rolled. Find P(both even).

Answer: **Verification:**

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Problem 7: Marbles

2 red, 8 blue. Without replacement, find $P(\text{both red})$.

Answer: **Verification:**

Problem 8: With replacement

2 red, 8 blue. With replacement, find $P(\text{both red})$.

Answer: **Verification:**

Problem 9: Coins

Three fair coins. Find $P(HHH)$.

Answer: **Verification:**

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Problem 10: Tree diagram

Why is a tree diagram useful for sequential (multi-stage) events?

Answer: **Verification:**

Problem 11: Cards

Without replacement, find P(ace then king).

Answer: **Verification:**

Problem 12: Signal words

The phrase "without replacement" usually signals which type of events?

Answer: **Verification:**

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Problem 13: Multiply rule

Independent events: $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{2}$. Find $P(A \text{ and } B)$.

Answer: **Verification:**

Problem 14: Three-stage

Three marbles are drawn without replacement from a bag of 3 red and 5 blue. Find $P(\text{all three red})$.

Answer: **Verification:**

Problem 15: Challenge

A bag has 6 white and 4 black balls. Three balls are drawn without replacement. Find $P(\text{exactly the order white, white, black})$, as a fraction in lowest terms.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

✓ Answer Key

1. $\frac{1}{4}$
2. $\frac{1}{4}$
3. $\frac{2}{9}$
4. Independent
5. $\frac{1}{17}$
6. $\frac{1}{4}$
7. $\frac{1}{45}$
8. $\frac{1}{25}$
9. $\frac{1}{8}$
10. It organizes every branching outcome and its probability, making them easy to multiply and add
11. $\frac{4}{663}$
12. Dependent
13. $\frac{1}{6}$
14. $\frac{1}{56}$
15. $\frac{1}{6}$

Great work — keep practising!